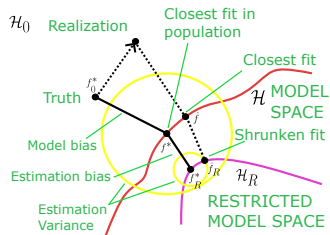
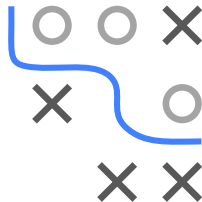


Introduction to Machine Learning

Regularization

Bias-variance Tradeoff



Learning goals

- Understand the bias-variance trade-off
- Know the definition of model bias, estimation bias, and estimation variance

BIAS-VARIANCE TRADEOFF I

In this slide set, we will visualize the bias-variance trade-off.

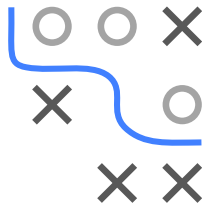
We consider a DGP $\mathbb{P}_{xy}$ with $\mathcal{Y} \subset \mathbb{R}$ and the L2 loss L . We measure the distance between models $f : \mathcal{X} \rightarrow \mathbb{R}^g$ via

$$d(f, f') = \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\mathbf{x}}} [L(f(\mathbf{x}), f'(\mathbf{x}))].$$

We define f_0^* as the risk minimizer such that

$$f_0^* \in \arg \min_{f \in \mathcal{H}_0} \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{xy}} [L(y, f(\mathbf{x}))]$$

where $\mathcal{H}_0 = \{f : \mathcal{X} \rightarrow \mathbb{R} \mid d(\underline{0}, f) < \infty\}$ and $\underline{0} : \mathcal{X} \rightarrow \{0\}$.



BIAS-VARIANCE TRADEOFF II

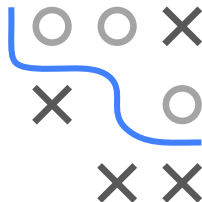
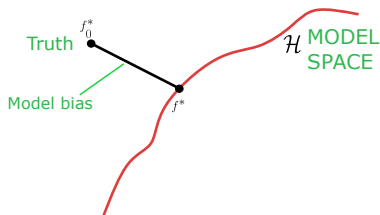
Our model space $\mathcal{H}$ usually is a proper subset of $\mathcal{H}_0$ and in general $f_0^* \notin \mathcal{H}$.

We define f^* as the risk minimizer in $\mathcal{H}$, i.e.,

$$f^* \in \arg \min_{f \in \mathcal{H}} \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{xy}} [L(f(\mathbf{x}, y))].$$

$f^* \in \mathcal{H}$ is closest to f_0^* , and we call $d(f_0^*, f^*)$ the model bias.

$\mathcal{H}_0$

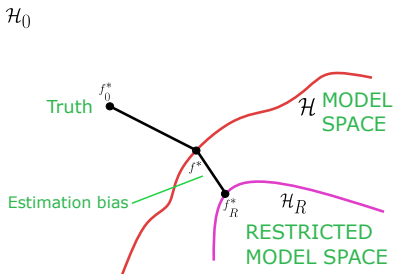
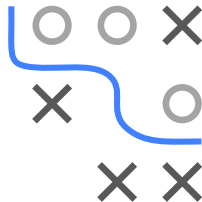


BIAS-VARIANCE TRADEOFF III

By regularizing our model, we further restrict the model space so that $\mathcal{H}_R$ is a proper subset of $\mathcal{H}$. We define f_R^* as the risk minimizer in $\mathcal{H}_R$, i.e.,

$$f_R^* \in \arg \min_{f \in \mathcal{H}_R} \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{xy}} [L(f(\mathbf{x}, y))].$$

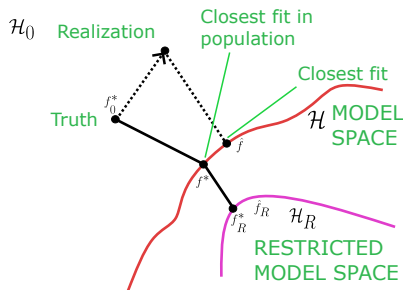
$f_R^* \in \mathcal{H}_R$ is closest to f_{true} , and we call $d(f_R^*, f^*)$ the estimation bias.



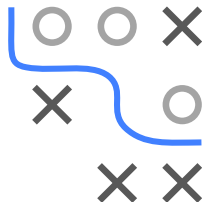
BIAS-VARIANCE TRADEOFF IV

We sample a finite dataset $\mathcal{D} = (\mathbf{x}^{(i)}, y^{(i)})^n \in (\mathbb{P}_{xy})^n$ and find via ERM

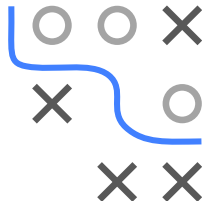
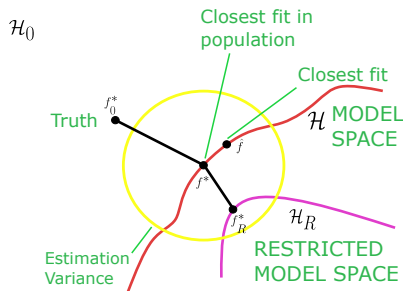
$$\hat{f} \in \arg \min_{f \in \mathcal{H}} \sum_{i=1}^n L(y^{(i)}, \hat{f}(\mathbf{x}^{(i)})) .$$



Note that the realization is only shown in the visualization for didactic purposes but is not an element of $\mathcal{H}_0$.



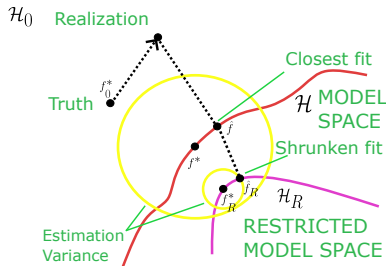
BIAS-VARIANCE TRADEOFF V



BIAS-VARIANCE TRADEOFF VI

We repeat the previous construction in the restricted model space $\mathcal{H}_R$ and sample $\hat{f}_R$ such that

$$\hat{f}_R \in \arg \min_{f \in \mathcal{H}_R} \sum_{i=1}^n L(y^{(i)}, \hat{f}(\mathbf{x}^{(i)})) .$$



- We can measure the spread of sampled $\hat{f}_R$ around f_R^* via $\delta = \text{Var}_{\mathcal{D}} [d(f_R^*, \hat{f}_R)]$ which we also call estimation variance.
- We observe that the increased bias results in a smaller estimation variance in $\mathcal{H}_R$ compared to $\mathcal{H}$.

